

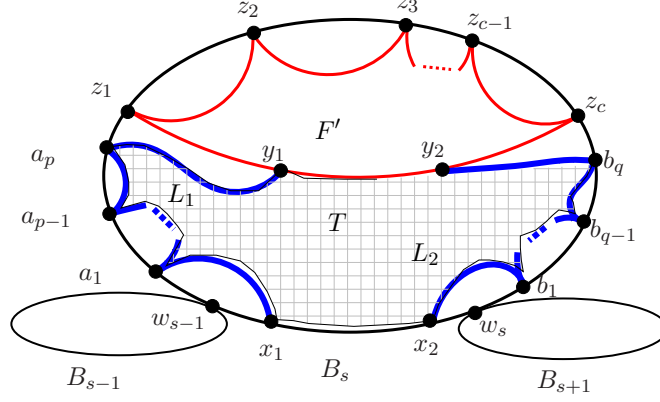


Figure 3: Some notations on B_s .

a contradiction. If $|G_2| \geq t$, then $|B_r| = |G_2| \geq t$ and $m(B_r) < m(G_2) = m(G)$. Since B_r is 2-connected and

$$m(B_r) \leq m(G) - 1 < \frac{n - (t - 1)}{3t - 7} - 1 = \frac{n - (3t - 7) - (t - 1)}{3t - 7} < \frac{|B_r| - (t - 1)}{3t - 7},$$

contradicting the choice of G . $\square$

4 Proof of Theorem 1.5

The proof proceeds by induction. If $k^{\log_2 3} \leq n \leq 36k^{\log_2 3}$, then

$$e(G) \leq 3n - 6 \leq 3n - 6 - \frac{n}{36k^{1+\log_2 3}} + \frac{k^3}{4},$$

the result holds. So, we assume that

$$n > 36k^{\log_2 3}. \quad (2)$$

Let G be a $2C_k$ -free plane graph with $e(G) = \exp(n, 2C_k)$. It is clear that G is connected; otherwise we can add an edge between two components of G to make it remain $2C_k$ -free, a contradiction. If G is C_k -free, then by Theorem 1.3, $e(G) \leq 3n - 6 - \frac{n}{4k^{\log_2 3}} \leq 3n - 6 - \frac{n}{36k^{1+\log_2 3}} + \frac{k^2}{2}$, the result holds. So, we assume that G contains a C_k .

Case 1. G is 2-connected.

In that case, the boundary of each face in G is a cycle. Assume that c is the smallest number of vertices that need to be removed from G to make it C_k -free, and let $v_1, v_2, \dots, v_c$ be one such set of vertices. Then $c \leq k$, since we can remove vertices of a k -cycle to make it C_k -free. Moreover, for any $i, j \in [c]$, there exists a k -cycle in G such that v_i is contained in the k -cycle, but v_j not.

Case 2.1. $c = 1$.

Since G is 2-connected, it follows that $G' = G - v_1$ is connected and is C_k -free. Assume that G' has r blocks $B_1, B_2, \dots, B_r$, where $|B_i| \geq k^{\log_2 3}$ for $i \in [q]$ and $|B_i| < k^{\log_2 3}$ for $q + 1 \leq i \leq r$. For each $i \in [q]$, since B_i is a C_k -free 2-connected plane graph, it follows that

$$m(B_i) \geq \frac{|B_i| - (k^{\log_2 3} - 1)}{3k^{\log_2 3} - 7};$$